Efrén López-Morales

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Professional Appointments

2025–Present **Assistant Professor**, *Computer Science*, New Mexico State University, Las Cruces, NM

Education

2021–2025 **Ph.D. in Computer Science**, *Texas A&M University-Corpus Christi*
Advisor: Carlos Rubio-Medrano
Thesis: *CTI-for-CPS*: Securing Cyber-Physical Systems via Advanced Cyber Threat Intelligence Methods

2018–2020 **Master's Degree in Computer Science**, *Arizona State University*
Advisors: Gail-Joon Ahn, Adam Doupé, and Carlos Rubio-Medrano
Thesis: HoneyPLC: A Next-generation Honeypot for Industrial Control Systems

2009–2012 **Bachelor's Degree in Computer Networks**, *Universidad Autónoma de Guadalajara*

Publications

Efrén López-Morales*, Ulysse Planta*, Gabriele Marra, Carlos González, Jacob Hopkins, Majid Garoosi, Elias Obreque, Carlos Rubio-Medrano, and Ali Abbasi, "HoneySat: A Network-based Satellite Honeypot Framework," In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, San Diego, CA, USA, February 2026.
*Co-first authors

Fangzhou Dong, Arvind S Raj, **Efrén López-Morales**, Siyu Liu, Yan Shoshitaishvili, Tiffany Bao, Adam Doupé, Muslum Ozgur Ozmen, and Ruoyu Wang, "Discovering Blind-Trust Vulnerabilities in PLC Binaries via State Machine Recovery," In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, San Diego, CA, USA, February 2026.

Efrén López-Morales, "Orbital Escalation: Modeling Satellite Ransomware Attacks Using Game Theory," In *Proceedings of the Workshop on the Security of Space and Satellite Systems (SpaceSec)*, San Diego, CA, USA, February 2026.

Efrén López-Morales, Ulysse Planta, Carlos E. Rubio-Medrano, Ali Abbasi, Alvaro Cardenas, "SoK: Security of Programmable Logic Controllers," *In Proceedings of the 33rd USENIX Security Symposium*, Philadelphia, PA, USA, August 2024.

Bhupendra Acharya, Dario Lazzaro, **Efrén López-Morales**, Adam Oest, Muhammad Saad, Antonio Emanuele Cinà, Lea Schönherr, Thorsten Holz, "The Imitation Game: Exploring Brand Impersonation Attacks on Social Media Platforms," *In Proceedings of the 33rd USENIX Security Symposium*, Philadelphia, PA, USA, August 2024.

Efrén López-Morales, Jacob Hopkins, Alvaro Cardenas, Ali Abbasi, Carlos E. Rubio-Medrano, "By the Numbers: Towards Standard Evaluation Metrics for Programmable Logic Controllers' Defenses," *In Proceedings of the 2nd International Workshop on Re-design Industrial Control Systems with Security (RICSS)*, Salt Lake City, UT, USA, October 2024.

Luis Salazar, **Efrén López-Morales**, Juan Lozano, Carlos E. Rubio-Medrano, Alvaro Cardenas, "ICSNet: A Hybrid-Interaction Honeynet for Industrial Control Systems," *In Proceedings of the 6th Joint Workshop on CPS & IoT Security and Privacy (CPSIoTSec)*, Salt Lake City, UT, USA, October 2024.

Efrén López-Morales, Carlos E. Rubio-Medrano, Adam Doupé, Yan Shoshitaishvili, Ruoyu Wang, Tiffany Bao, and Gail-Joon Ahn, "HoneyPLC: A Next-generation Honeypot for Industrial Control Systems," *In: Bao, T., Tambe, M., Wang, C. (eds) Cyber Deception. Advances in Information Security, vol 89. Springer, Cham*, October, 2022.

Efrén López-Morales, Carlos E. Rubio-Medrano, Adam Doupé, Yan Shoshitaishvili, Ruoyu Wang, Tiffany Bao, and Gail-Joon Ahn, "HoneyPLC: A Next-generation Honeypot for Industrial Control Systems," *In Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security 2020*, Online, November 2020.

Honors & Awards

Feb 2025 **Cyber-Physical Systems (CPS) Rising Star**, National Science Foundation

Nov 2024 **CAHSI Student Scholar**, Computing Alliance of Hispanic Serving Institutions

Sep 2023 **Google Conference Scholarship**

Oct 2021 **CONACYT Doctoral Scholarship**, Mexican National Council for Science and Technology

Apr 2021 **SAGE Fellowship**, Texas A&M University-Corpus Christi

2018 **Fulbright-García Robles Scholarship**, Fulbright COMEXUS

Invited Talks

Mar 2026 **Preparing Your Thesis Proposal and Becoming a Ph.D. Candidate**, Grad Cohort Workshop Southwest, Computing Research Association (CRA) (Online)

Apr 2024 **Design and Implementation of a Satellite Honeypot**, 1st Internet of Space Things (IoST) Workshop, University of Chile (Online)

Mar 2022 **Ciberseguridad: Habilidades y Oportunidades**, Universidad de la Sierra Juárez (Oaxaca, Mexico)

Service

Program Committee

2026 IEEE Conference on Resilience and Integrated Security for Space & Critical Systems (RISC)

2026 ACM Conference on Computer and Communications Security (CCS)

Conference Reviewer

2025 ACM Conference on Data and Application Security and Privacy (CODASPY)

Journal Reviewer

2026 ACM Transactions on Privacy and Security

2025 IEEE Transactions on Industrial Informatics

2025 IEEE Security and Privacy

2025 IEEE Transactions on Dependable and Secure Computing

2023 IEEE Transactions on Dependable and Secure Computing

Teaching Experience

Instructor

Spring 2026 CSCI 4205/5205 "Computer Security", *New Mexico State University*

Fall 2025 CSCI 4240/5240 "Software Reverse Engineering", *New Mexico State University*

Fall 2023 COSC 4367 "Firewalls and Intrusion Detection Systems", *Texas A&M University-Corpus Christi*

Research Experience

Summer 2023 **Research Internship**, *CISPA Helmholtz Center for Information Security*
Advisors: Ali Abbasi and Thorsten Holz

Summer 2022 **Visiting Researcher**, *University of California, Santa Cruz*
Advisor: Álvaro Cárdenas

In the Media

Feb 27, 2026 **"Hackers Tricked by Fake Satellite in Groundbreaking Cybersecurity Sting"**, featured in Orbital Today

Mar 27, 2025 **"Security Research Gaps Leave Critical Infrastructure Open to Cyberattack"**, featured in Communications of the ACM Magazine

Industry Experience

2020–2021 **Network Support Engineer**, *Cisco Meraki*, Diagnosed and troubleshooted routing and switching, wireless, security, SD-WAN and firewalls

Languages

Spanish: Native

English: Proficient

Brazilian Portuguese: Proficient

German: Elementary proficiency

Last updated on March 15, 2026